A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 16

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Deaths rise as smoke concentration rises, peaking at about 930 per day on 8 Dec when smoke was highest (4.5 mg m^{-3}) **1** Deaths fall as smoke falls but remain above the starting level for several days (time lag / delayed effect) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (930 - 250) / 250 x 100 **1** = 272% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 3 points below. **1 Anticyclonic conditions:** clear skies allowed rapid radiation of heat from the ground at night so the ground and air near it cooled **1** Low wind velocity so air layers did not mix – London lies in a valley / basin where cold dense air collected: a temperature inversion formed **1** Smoke from coal fires was trapped below the inversion – moist air cooled to its dew point forming fog – smoke + fog = smog – the smog's high albedo prolonged the inversion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **temperature inversion** → a layer of warm air above cooler air that traps pollutants • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Smoke particles (PM10) are inhaled deeply and irritate the lungs / kill cilia so particles and bacteria are not cleared, causing bronchitis / infections **1** These groups have weaker lungs / immune systems / less reserve so the extra stress (with sulfur dioxide acting synergistically) was fatal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **PM10** → fine solid particles / smoke and dust	2
01.5	1 mark for the point below. **1** Clean Air Act (1956) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1 Single:** 5.8 x 20 x 12 = 1392 W **1 Argon:** 1.6 x 20 x 12 = 384 W **1** Reduction = 1008 W (accept 1010) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The converse of any point, ie the same comparison written the other way round. • The value alone - the unit (W) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	3
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** A wider gap of trapped gas reduces conduction because gas is a poor conductor **1** If the gap is too wide convection currents start in the gas, carrying heat across – the optimum depends on the gas and window size **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Low-e coating reflects long-wavelength infrared back into the room, reducing radiation losses **1** Argon / krypton / xenon have lower thermal conductivity than air so less conduction across the gap **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Panes must be held apart by small spacers which cannot be long, so the gap is only about 0.2 mm **1** Used where there is not room for thick gas-filled multiple glazing (eg historic buildings) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.5	1 mark for the point below. **1** Each extra layer saves less energy (diminishing returns) while cost rises, so beyond the optimum the cost of insulation exceeds the value of energy saved **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-231; AQA spec 3.3.4.2.2.	
03.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Human = 120 + 60 + 30 = 210 – total = 325 **1** 210 / 325 x 100 = 64.6% (accept 65%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • A bare wrong answer with no working.	2
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Nitrogen and hydrogen react to form ammonia over an iron catalyst at high temperature and pressure – ammonia then converted to nitrates for fertiliser **1** Uses large amounts of energy from fossil fuels (also the hydrogen source), releasing CO2 and NOx **1** Biological fixation by bacteria occurs at low temperature using solar energy captured by plants (enzymes lower activation energy) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	3

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Eutrophication of water bodies by leached nitrates / sewage **1** Photochemical smogs and tropospheric ozone from NOx **1** Acid rain (nitric acid) **1** Nitrous oxide as a greenhouse gas **1** NOx toxicity / respiratory disease **1** Nitrate in drinking water (methaemoglobinaemia) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Eutrophication** → nutrient enrichment of water leading to algal growth • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Carbon is lost as CO2 (composting) or CH4 (digestion) so the C:N ratio improves / more nitrogen-rich **1** Bulk and odour reduced, pathogens killed, easier to apply – digestion also yields biogas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 154-155; AQA spec 3.2.4.3.	
04.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (4.8 - 1.4) / 4.8 x 100 **1** = 70.8% (accept 71%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** A tributary joins the river, more than doubling the flow (15 to 36 m^3 s^{-1}) **1** So the pollutant is diluted by the extra water (dispersal), not broken down **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** The pollutant is persistent so it is not degraded / broken down as it travels **1** Little extra water joins the river (flow 36 to 38) so little further dilution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	2
04.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Lake has a long residence time / water does not flow away so the pollutant accumulates rather than being carried away **1** Smaller volume of water / less dilution / less dispersal by currents **1** Persistent pollutant gradually becomes more concentrated (eg by evaporation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime • **Persistent** → does not break down / stays in the environment for a long time	2
04.5	**1 mark each** 1 mark for each of the 2 points below. **1 Marine discharge:** large volume of water and currents dilute and disperse the pollutant to acceptably low concentrations (control of emission location) **1 River:** small volume so concentrations stay high / drinking water abstraction / sensitive wildlife downstream **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abstraction** → water taken out of a river, lake or aquifer	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-241, 254-255; AQA spec 3.4.1, 3.4.2.1.	
05.1	**2 marks** 2 marks for the point below; award fewer where it is only partly made. **2** When there is a difference in level between the sea and the basin: on the ebb (about 6-8 hours) water flows out through the turbines – no generation when levels are equal (slack water) **OR** possible on the flood when the sea is higher than the basin **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Two periods of slack water in each 12.5-hour cycle (twice in about 25 hours) give no flow – the timing does not match demand **1** Tides depend on the positions of the Moon and Sun which are known, so output can be forecast years ahead and alternative supplies planned **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1 Spring tides:** Sun and Moon aligned so their gravitational effects combine giving a large tidal range – neap: effects oppose giving a small range **1** Energy depends on the mass of water and height difference, so output is much greater at springs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** When demand is low, surplus electricity pumps water into the lagoon above sea level **1** The stored water is released through the turbines later when demand is high (pumped storage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pumped storage** → water pumped uphill when demand is low and released to generate at peak	2
05.5	**1 mark each** 1 mark for each of the 2 points below. **1** Large tidal range (eg Severn Estuary) **1** Estuaries, bays or headlands / channels that focus the flow and increase velocity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 214-215; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** Surplus electricity (eg windy nights) electrolyses water to produce hydrogen which is stored **1** Hydrogen has a high energy density so it can be transported / stored for long periods **1** Released by combustion for heat, industrial processes or steam turbines **OR** by fuel cells combining hydrogen and oxygen electrochemically to produce electricity and water **1** Can also be used as a vehicle fuel or converted to methane / ammonia **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4
06.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Compressed gas (up to 700 times normal volume) - energy to run the compressor, strong tanks **1** Liquefied at very low temperature - energy for refrigeration and compression **1** Adsorbed on a metal matrix (metal hydride) at low pressure - tanks larger and heavier than a petrol tank storing the same energy **1** Conversion to ammonia - some energy needed but less than the alternatives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Both convert chemical energy to electricity electrochemically, but a battery stores a fixed amount of chemicals which are reformed by recharging **1** A fuel cell keeps producing electricity as long as fresh fuel (hydrogen / alcohol) and oxidant are supplied **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **fuel cell** → a cell that produces electricity from hydrogen and oxygen	2

Do not write outside the box

Question	Marking guidance	Marks
06.4	1 mark for the point below. **1** Using hydrogen made from surplus renewable electricity as the main store and carrier of energy, released when demand exceeds supply or intermittent sources fail, for heating, vehicles and electricity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 219-221; AQA spec 3.3.4.1.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Natural pressure of gas above the oil (or dissolved in it) or water beneath forces oil up the production well – pump-jacks ('nodding donkeys') increase flow **1** Pressure falls as oil is removed so the remaining oil no longer flows to the well **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Water or natural gas is pumped down an injection well to maintain the pressure and flow of oil (recovery to about 40%) **1** Captured CO2 can be pumped down instead, increasing oil recovery while storing the CO2 underground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carbon capture** → capturing carbon dioxide from flue gases and storing it underground	2
07.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Steam injection (from combustion or solar parabolic concentrators) to heat the oil and reduce its viscosity **1** Controlled underground combustion to heat the oil **1** Detergents / solvents to reduce surface tension so oil flows more easily **1** Bacteria that partially digest heavy oil into lighter oils and produce CO2 to maintain pressure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
07.4	**1 mark each** 1 mark for each of the 3 points below. **1** Many wells can be drilled from a single platform, reducing land take / number of rigs **1** Deposits under locations where rigs cannot be placed (urban areas, sensitive habitats) can be reached **1** Drilling can follow softer strata (quicker) and target multiple small reservoirs up to 10 km away, increasing total recovery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187-188; AQA spec 3.3.4.1.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** Dissolved oxygen fluctuates daily / 24-hour cycle: highest in the afternoon (about 13-13.5 mg dm^{-3} at 16 and 40 h) and lowest at night / early morning (2.5-3.5 mg dm^{-3} at 4 and 28 h) **1** The night-time minimum is lower on the second night (2.5 vs 3.5) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.2	**1 mark each** 1 mark for each of the 4 points below. **1** Nitrates (from fertiliser runoff) and phosphates (from sewage effluent) cause eutrophication: rapid growth of algae / cyanobacteria (algal bloom) **1** During the day photosynthesis by the dense algae releases oxygen so concentrations rise (above the normal saturation level) **1** At night there is no photosynthesis but respiration by algae / animals / bacteria continues, using oxygen **1** Aerobic bacterial decomposition of dead algae / shaded macrophytes uses oxygen – warm summer water holds less oxygen and decay is rapid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	4

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** At night no photosynthesis replaces the oxygen used in respiration / decomposition so concentrations fall below the minimum fish can tolerate **1** At the end of summer large amounts of dead vegetation decay rapidly in warm water causing severe deoxygenation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water	2
08.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Cyanobacteria release toxins harmful to livestock, pets and people **1** Floating algae shade submerged / rooted macrophytes which die, disrupting food webs **1** Algae are an unreliable food source (die off rapidly) so food webs are poor **1** Blocked filters / unpleasant taste and smell in water supplies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-268; AQA spec 3.4.3.2.9.	
09.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** The readings are precise but may not be accurate **Also accept** • **precise** → readings close together / fine resolution • **accurate** → close to the true value	1
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** A result that differs from other results with which it was expected to be similar **1 Repeat the study / measurement:** if it recurs it is a 'real' result caused by another variable – if not it was an error and can be excluded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** To test the reliability of the method / equipment **1** To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Stages of scientific method illustrated by ozone: identification (Antarctic UV data), hypothesis (Rowland-Molina 1974), data collection / analysis (Dobson units, satellites, balloons confirming ClO), proposed solutions, enactment (Vienna 1985, Montreal 1987) • **Evidence of success:** ozone hole area and depth stabilised and beginning to shrink – CFC concentrations in the atmosphere falling – UV levels not rising further – global ozone loss limited to about 4% • **Reasons for success:** international recognition of serious consequences (skin cancer) – agreement of nearly every country – legally binding targets with a fund for developing countries – relatively few producers / products – alternatives (HCFCs, HFCs, pump sprays, hydrocarbons) already available and cheap – no lifestyle change needed • **Limits:** persistence of CFCs means recovery takes decades – some ODSs still used – HFC replacements are greenhouse gases • **Climate change contrast:** greenhouse gases come from almost every human activity (energy, transport, agriculture, industry) in every country – fossil fuels underpin economies so control requires major lifestyle and economic change • Agreements (Kyoto, Paris) have weaker / non-binding targets, disputes between developed and developing nations – emissions still rising – natural fluctuations, time delays and tipping points make cause-effect less clear to the public	9

Do not write outside the box

Question	Marking guidance	Marks
	• **Judgement:** ozone control was a technical substitution problem – climate change is a systemic one, so the ozone success is only partly transferable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • The converse of any point, ie the same comparison written the other way round. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **persistence** → does not break down / stays in the environment for a long time • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	
	Model answer - Level 3 (9 marks)	
	Ozone depletion followed the scientific method almost textbook-fashion: Antarctic UV and ozone measurements identified a problem, Rowland and Molina's 1974 hypothesis explained it, Dobson-unit, satellite and balloon data confirmed chlorine monoxide in the stratosphere, and the Montreal Protocol enacted a solution. It has largely worked: CFC concentrations are falling, the ozone hole has stabilised and is starting to shrink, and global ozone loss was limited to about 4%.	
	Success came from particular conditions. The consequence - skin cancer - was clear and frightening, nearly every country agreed, targets were legally binding with a fund to help developing countries, there were relatively few producers and products, and alternatives (HCFCs, HFCs, hydrocarbons, pump sprays) already existed, so substitution cost little. Even so recovery will take decades because CFCs are so persistent, some ozone-depleting substances are still in use, and the HFC replacements are themselves greenhouse gases.	
	Climate change has none of these advantages. Greenhouse gases come from almost every activity - energy, transport, agriculture, industry - in every country, and fossil fuels underpin entire economies, so control means changing lifestyles and economies rather than swapping one chemical for another. Kyoto and Paris set weaker, largely non-binding targets, developed and developing nations dispute responsibility, and natural fluctuations, time lags and tipping points make cause and effect less obvious to the public. Emissions are still rising. The ozone story shows international agreement can work, but it was a technical substitution problem; climate change is systemic, so its lessons transfer only partly.	
	Weaker answer - Level 2 (4 marks)	
	Ozone depletion was caused by CFCs. Scientists found the hole over Antarctica and the Montreal Protocol banned CFCs, so the ozone layer is now recovering. This was successful because everyone agreed and there were alternatives to CFCs. Climate change has not been tackled as well. There have been agreements like Kyoto and Paris but carbon dioxide is still going up. This is because fossil fuels are used everywhere and countries do not agree. So ozone depletion was solved better than climate change.	

Do not write outside the box

Question	Marking guidance	Marks
	What makes the difference	
	• Level 2 gives the right outline but Level 3 supports it with evidence (falling CFC levels, 4% loss, Dobson units, ClO detection) and explains each reason for the difference.	
	• Level 3 acknowledges the limits of the ozone success (persistence, HFCs) - balance is expected at the top level.	
	• The final judgement in Level 3 characterises the difference (substitution vs systemic problem); Level 2 just restates the question.	
10.2	1 mark for the point below. **1** Vienna Convention for the Protection of the Ozone Layer - no legally binding goals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Ozone Layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113, 105-106; AQA spec 3.2.1.3, 3.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Features of fertile soil:** water, nutrients, air, organic matter, pH, biota, texture, structure, depth – slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • **Causes of degradation:** vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • **Types of erosion:** rain splash, runoff, wind, slumping – consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • **Management:** long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy	25
	• Evaluation and judgement on which strategies are most effective in different environments *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Nature of the problem:** interconnected atmosphere-hydrosphere-biosphere systems – changes over different time and spatial scales – natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change – time delay between cause and effect (ocean heat capacity) • **Limitations of historic data:** sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records • **Proxy data and their reliability:** dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems) • **Modern monitoring:** long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level • Feedback mechanisms and tipping points adding uncertainty – ecological uncertainty (species survival, distribution, fragmentation) • **Computer models:** how they are tested and refined – importance of accurate representative data – scenario-based predictions and ranges	25
	• Uncertainty over ENSO / Conveyor / jet stream responses • **Evaluation and conclusion:** confidence in the overall trend and human cause despite detailed uncertainty – role of peer review and international bodies (IPCC) *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3.	

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